Theoretical Architecture, Axiomatic Foundations, and Formal Verification of the Bekenstein–Nyquist Shock Worldline Conjecture

Bekenstein–Nyquist Research Program

October 2026

Abstract

The Bekenstein–Nyquist Shock Worldline Conjecture (BNSWC) establishes a rigorous mathematical bridge between relativistic information density bounds, microlocal partial differential equation (PDE) theory, and Boolean or unitary circuit complexity. The conjecture is founded on three physical premises—the Bekenstein spatial information bound (Λ_{Bek}), the Margolus–Levitin rate limit on local state changes (ν_{ML}), and reflectionless local unitarity ($U^\dagger U = I$)—formalized as four minimal axioms (Axioms A1–A4) under scale-generating nonlinearity ($[P_\Lambda, U_{\mathrm{nl}}] \neq 0$). Smooth, sub-Nyquist field configurations evolve via spatially parallel, constant-depth updates ($\mathcal{O}(1)$ circuit depth), whereas nonlinear convective steepening at a hydrodynamic or field-theoretic shock front saturates the local spatial bandwidth. Rather than postulating auxiliary degrees of freedom by hand, we prove via Stinespring/Sz.-Nagy dilation (`UnitaryBandwidthDepth.lean`) that preserving local unitarity without violating spatial bandwidth bounds mathematically forces the existence of an orthogonal worldline clock fiber $\mathcal{H}_{\mathrm{clock}} = V_\Lambda^\perp$ of rank at least $r = \mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$, a 90° microlocal signature rotation in the complexified cotangent bundle across the saturation horizon $\Sigma(x,t) = 0$, and an effective Landau–Ginzburg–Higgs potential $V_{\mathrm{eff}}(c) = \Sigma_0|c|^2 + \lambda|c|^4$. This monograph presents the unified axiomatic architecture (including the equivalence between the Emergent Stinespring Dilation formulation and the Primitive Bipartite Fiber-Bundle formulation), the five-phase analytical proofs, the machine-checked Lean 4 verification, the numerical verification across four progressive hydrodynamic lattice benchmarks, the holographic Ryu–Takayanagi bulk-extrusion correspondence, and the formal duality with the Landau–Ginzburg–Higgs mechanism.

Contents

Executive Synthesis: The Three Physical Premises and Deductive Architecture **2**

1 Axiomatic Foundations, Deductive Structure, and the Aliasing–Unitarity No-Go Lemma (Phase 1) **3**
- 1.1 The Minimal Four-Axiom Formulation (Emergent Dilation Architecture) 3
- 1.2 Deductive Emergence of the Clock Fiber, 90° Rotor, and Higgs Potential 4
- 1.3 Axiomatic Equivalence: Emergent Dilation vs. Primitive Bipartite Bundle 5
- 1.4 Analytical Proof Architecture for the Nyquist–Lax Trichotomy 6

2 Microlocal Analysis of the 90° Causal Rotation (Phase 2) **7**
- 2.1 The Bekenstein–Nyquist Saturation Functional and Mixed-Type PDEs 7
- 2.2 Complex Coordinate Continuation and Internal Clock Winding 8

3 Complexity Theory and Sequential Depth Lower Bounds (Phase 3) 8
3.1 The Sub-Stepping Depth Theorem and Holographic Shock Complexity 8
3.2 P-Completeness of Shock Worldline Networks and the Monotone Planar Circuit Trap 9

4 Computational Verification on Bandlimited Hydrodynamic Lattices (Phase 4) 9
4.1 The $1+1$D Unitary Nyquist Rotor Lattice (Experiment H.3 / Milestone 2) . . . 9
4.2 Comparative Shock-Capturing Benchmarks and Poisson–Euler RG Duality 10
4.3 Spacetime Causal DAG Critical-Path Scaling and Asymptotic Tightness (Experiment H.4 / Milestone 3) 10
4.4 The Bandwidth–Saturation Equivalence Principle and Holographic Bulk Extrusion 11
4.4.1 Holographic Bulk Extrusion, Ryu–Takayanagi Min-Cut Wire Counting, and Effective g_{00} Horizon Geometry 11
4.5 Two-Shock Collision Interferometry, Coherent Clock-Mass Annihilation, and the Ginzburg–Landau–Higgs Duality (Milestone 4 & Phase 3.2) 12
4.5.1 Evasion of the Monotone Planar Circuit Trap via Orthogonal Two-Mode Haar Clock Readback 12
4.5.2 Experimental Verification: Gate Truth Table, Coalescence Kinematics, and Ramsey Interference Fringes 13
4.5.3 Duality Between the Bekenstein–Nyquist Clock Fiber and the Landau–Ginzburg–Higgs Mechanism 13

5 Machine-Checked Formal Verification in Lean 4 (Phase 5, Phase A & Phase B) 14
5.1 Core Bandwidth–Depth Tradeoff (`UnitaryBandwidthDepth.lean`) 14
5.2 Extended Physics Formalizations and Upstream `Physlib` / `QuantumInfo` Bridges 15

6 Integrated Synthesis 17

Executive Synthesis: The Three Physical Premises and Deductive Architecture

The Bekenstein–Nyquist Shock Worldline Conjecture rests on the simultaneous enforcement of three foundational principles of fundamental physics that have historically been analyzed in isolation:

1. **The Bekenstein Spatial Bandwidth Bound ($\Lambda_{\mathbf{Bek}}$):** General relativity and black hole thermodynamics impose the Bekenstein bound, restricting the maximum information $I_{\max}$ storable within a spherical region of radius R and energy E prior to Schwarzschild gravitational collapse to
$$I_{\max} \leq \frac{2\pi E R}{\hbar c \ln 2}. \tag{1}$$
Following Achim Kempf's covariant bandlimitation framework, this finite spatial information density imposes a hard ultraviolet wavenumber ceiling $|k| \leq \Lambda_{\mathrm{Bek}}$, restricting observable continuum fields to a Paley–Wiener Hilbert space $\mathcal{H}_\Lambda = PW_{\Lambda_{\mathrm{Bek}}}(\mathbb{R})$ obeying Bernstein's inequality $\|u^{(r)}\|_2 \leq \Lambda_{\mathrm{Bek}}^r \|u\|_2$. By the Shannon–Nyquist and Beurling–Landau sampling theorems, $\mathcal{H}_\Lambda$ is unitarily isomorphic to a discrete sampling lattice $\ell^2(\mathbb{Z}_{\Delta x})$ of fundamental spacing $\Delta x = \pi/\Lambda_{\mathrm{Bek}}$ without breaking continuous translational or Poincaré symmetries.

2. **The Margolus–Levitin Rate Limit on State Changes (ν_{ML}):** Quantum mechanics enforces the Margolus–Levitin theorem, which bounds the maximum frequency of distinguishable local transitions between orthogonal quantum states by

$$\nu_{\mathrm{ML}} \leq \frac{2E}{\pi\hbar}, \qquad \Delta\tau_{\mathrm{ML}} = \frac{\pi}{\nu_{\mathrm{ML}}} \geq \frac{\pi^2\hbar}{2E}. \tag{2}$$

   Together, the spatial bandwidth ceiling Λ_{Bek} and the temporal state-change rate limit ν_{ML} define a finite local spacetime information-processing capacity per unit length per unit time:

$$C_{\mathrm{self}} \equiv \theta_{\mathrm{Nyq}} = \Lambda_{\mathrm{Bek}}\nu_{\mathrm{ML}}. \tag{3}$$

3. **Local Microscopic Unitarity vs. Landauer Erasure ($U^\dagger U = I$):** Fundamental quantum evolution is strictly unitary and causally local (bounded spatial fan-in radius r_0 per parallel step), conserving total Hilbert-space norm and phase-space volume. While linear dispersive waves propagate trivially inside $\mathcal{H}_\Lambda$, nonlinear convective steepening (such as $u\partial_x u$) multiplies modes in the upper half of the Brillouin zone ($\Lambda_{\mathrm{Bek}}/2 < |k| \leq \Lambda_{\mathrm{Bek}}$), driving spectral flux $R_{\mathrm{self}}(x,t) = \|F_{tx}(x,t)\|_{\mathrm{HS}}$ toward the forbidden ultraviolet zone $(\Lambda_{\mathrm{Bek}}, 2\Lambda_{\mathrm{Bek}}]$. Standard continuum hydrodynamics stabilizes shocks via viscous dissipation (violating unitarity via Landauer bit erasure $E_{\min} = nk_BT\ln 2$), whereas naive unitary lattice truncation reflects ultraviolet modes back into the bulk via Umklapp aliasing (producing unphysical Gibbs ringing).

The central thesis of this monograph is that nature resolves nonlinear shock steepening under these three constraints without Landauer erasure or Gibbs aliasing by dynamically trading **spatial parallelism** for **sequential temporal circuit depth** along the lower-dimensional shock worldline.

1 Axiomatic Foundations, Deductive Structure, and the Aliasing–Unitarity No-Go Lemma (Phase 1)

1.1 The Minimal Four-Axiom Formulation (Emergent Dilation Architecture)

To separate fundamental physical resource inputs from derived mathematical structures, we formulate the conjecture on a composite Hilbert space $\mathcal{H}_{\mathrm{tot}}$ governed by four primitive axioms: two kinematic resource bounds (Bekenstein and Margolus–Levitin), one dynamical nonlinearity condition, and one local unitarity conservation law.

Axiom 1.1 (Spatial Bandlimitation — The Bekenstein–Kempf Projector (A1))**.** Observable spatial fields reside in a Paley–Wiener subspace $V_\Lambda = P_\Lambda\mathcal{H}_{\mathrm{tot}} = PW_{\Lambda_{\mathrm{Bek}}}(\mathbb{R})$ defined by an idempotent, self-adjoint ultraviolet wavenumber projector $P_\Lambda^2 = P_\Lambda = P_\Lambda^\dagger$ with hard spectral cutoff $|k| \leq \Lambda_{\mathrm{Bek}}$, unitarily isomorphic to the discrete sampling lattice $\ell^2(\mathbb{Z}_{\Delta x})$ with fundamental spacing $\Delta x = \pi/\Lambda_{\mathrm{Bek}}$.

Axiom 1.2 (Rate Limit on State Changes — The Margolus–Levitin Sequential Channel (A2))**.** Any local state transition coupling the bandlimited spatial subspace V_Λ to an orthogonal degree of freedom proceeds through elementary rank-1 quantum channels ($\mathrm{rank}((I - P_\Lambda)W_k P_\Lambda) \leq 1$) whose transition frequency is bounded above by the Margolus–Levitin limit $\nu_{\mathrm{ML}} \leq \frac{2E}{\pi\hbar}$ (imposing a minimum sequential clock tick $\Delta\tau_{\mathrm{ML}} = \pi/\nu_{\mathrm{ML}}$ per elementary gate W_k).

Axiom 1.3 (Scale-Generating Nonlinearity — Non-Commutativity (A3))**.** The microscopic generator of spatial wave transport is nonlinear (convective mode-multiplying), so for steepening

Table 1: Five-Phase Architecture of the Bekenstein–Nyquist Shock Worldline Program.

Program Phase	Core Objective	Mathematical Apparatus	Target Deliverable	Falsification Criterion
Phase 1: Axioms & Kinematics	State Minimal Axioms A1–A4 and prove incompatibility of spatial parallelism and unitarity at a shock	Paley–Wiener spaces $\mathcal{H}_\Lambda$, discrete Lax pairs (L, M), Hilbert–Schmidt commutator norms	Minimal Axiomatization & **Nyquist–Lax Trichotomy**	Local bounded-fan-in spatial stencil resolving a Lax spike without dissipation or aliasing
Phase 2: Microlocal Dynamics	Derive 90° phase-space rotation converting UV spatial modes into temporal rotors	Mixed-type Tricomi–Keldysh PDEs, complexified cotangent bundle $T^*\mathcal{M}_\mathbb{C}$, Stokes connection	Signature flip from elliptic $(+,+)$ to hyperbolic $(-,+)$ across $\Sigma(x,t)=0$	Failure of complex coordinate continuation to preserve symplectic/unitary norm
Phase 3: Complexity Theory	Translate hyperbolic shock lightcones into sequential circuit depth lower bounds	Bounded-depth circuits (NC vs. P), DAG critical-path analysis, communication complexity	**Sub-Stepping Depth Theorem** and P-completeness of shock networks	Parallel prefix or spectral algorithms fast-forwarding intersecting shocks in $\mathcal{O}(\log^k N)$ depth
Phase 4: Computational Sandboxes	Benchmark unitary temporal sub-stepping against dissipative shock-capturing schemes	Bandlimited spectral lattices, unitary Haar–Givens rotors, co-moving clock registers in PyTorch	**Milestones 2, 3, & 4**: Zero-loss shock propagation, linear DAG scaling, coherent clock readback	Secular energy drift or non-physical dispersive reflection in temporal phase buffers
Phase 5: Formal Verification	Machine-check the finite-dimensional algebraic core of the bandwidth–depth tradeoff	Lean 4, `Matrix.Rank`, `UnitaryGroup`, block-partitioned sum types	Verified module `UnitaryBandwidthDepth.lean` proving $D \geq \operatorname{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$	Failure of commutator rank bounds to enforce a strict linear lower bound on sequential gate depth

wave profiles the unconstrained short-time propagator $U_{\mathrm{nl}} = \exp(\Delta t M(u))$ fails to commute with the spatial bandlimit projector P_Λ:

$$[P_\Lambda, U_{\mathrm{nl}}] \neq 0 \quad \Longleftrightarrow \quad \mathcal{A}_\Lambda[u] \equiv \left\| \left[P_\Lambda, [M(u), L(u)]\right] \right\|_{\mathrm{HS}} > 0, \tag{4}$$

where $(L(u), M(u))$ denotes the Lax pair of the nonlinear flow and $F_{tx}[u] = \partial_t M - \partial_x L + [M, L]$ is the zero-curvature defect.

Axiom 1.4 (Reflectionless Local Unitarity — No Erasure and No Aliasing (A4))**.** Time evolution on $\mathcal{H}_{\mathrm{tot}}$ is generated by bounded-fan-in local unitary operators ($U^\dagger U = I$, forbidding dissipative Landauer bit erasure $E_{\min} = nk_B T \ln 2$) that satisfy reflectionless impedance matching at the Brillouin boundary $|k| = \Lambda_{\mathrm{Bek}}$ (forbidding unphysical Umklapp spectral folding $k \mapsto k - 2\Lambda_{\mathrm{Bek}}$ or backward dispersive Gibbs reflections into V_Λ).

1.2 Deductive Emergence of the Clock Fiber, 90° Rotor, and Higgs Potential

Under Axioms A1–A4, the auxiliary structures of the Bekenstein–Nyquist framework are not assumed ad hoc; they emerge as rigorous mathematical deductions along a single linear chain:

$$\underbrace{\text{Axioms A1 + A3}}_{\text{Bandlimit + Nonlinearity}} \implies \underbrace{[P_\Lambda, U_{\mathrm{nl}}] \neq 0}_{\text{Phase 1 Defect}} \xrightarrow{+\text{A4}} \underbrace{V_\Lambda^\perp \equiv \mathcal{H}_{\mathrm{clock}}}_{\text{Phase 5 Clock Fiber}} \xrightarrow{+\text{A2}} \underbrace{\mathcal{D}(\gamma)\ \&\ \Sigma(x,t) < 0}_{\text{Phases 2–4 Rotor \& Higgs}} . \tag{5}$$

Specifically:

1. **Deduction 1 — Existence of the Orthogonal Worldline Clock Fiber $\mathcal{H}_{\text{clock}}$ (From A1 + A3 + A4):** By Axiom A3 ($[P_\Lambda, U_{\text{nl}}] \neq 0$), the projected spatial update $\Phi_{\Delta t} = P_\Lambda U_{\text{nl}} P_\Lambda$ satisfies $(P_\Lambda U_{\text{nl}} P_\Lambda)^\dagger (P_\Lambda U_{\text{nl}} P_\Lambda) = P_\Lambda - A^\dagger A$ with inter-band leakage $A = (I - P_\Lambda) U_{\text{nl}} P_\Lambda \neq 0$, making $\Phi_{\Delta t}$ a strict contraction on V_Λ. By Axiom A4 (Unitarity) and the Sz.-Nagy/Stinespring dilation theorem (machine-checked in Phase 5, `UnitaryBandwidthDepth.lean`), restoring unitarity mathematically requires an orthogonal complement $V_\Lambda^\perp = (I - P_\Lambda)\mathcal{H}_{\text{tot}} \equiv \mathcal{H}_{\text{clock}}$ of dimension at least $r = \text{rank}(A) = \frac{1}{2}\text{rank}([P_\Lambda, U_{\text{nl}}])$.

2. **Deduction 2 — Sequential Depth Lower Bound $\mathcal{D}(\gamma)$ (From Deduction 1 + A2):** Combining the commutator rank identity $\text{rank}([P_\Lambda, U_{\text{nl}}]) = 2r$ with Axiom A2 (each elementary gate couples at most a 1D clock wire at rate $\leq \nu_{\text{ML}}$) forces any unitary circuit realization $U_D = W_D \cdots W_1$ to have sequential depth $D \geq r = \frac{1}{2}\text{rank}([P_\Lambda, U_{\text{nl}}])$, yielding the continuum Sub-Stepping Depth Theorem along the shock worldline γ (Phase 3).

3. **Deduction 3 — Normalized Feasibility Margin $\Sigma(x,t)$ and the 90° Signature Flip (From A1–A4):** Identifying the local parallel channel capacity $C_{\text{self}} = \Lambda_{\text{Bek}}\nu_{\text{ML}}$ (from A1 and A2) and the nonlinear spectral spillover flux $R_{\text{self}}(x,t) = \|F_{tx}(x,t)\|_{\text{HS}}$ (from A3) defines the dimensionless channel feasibility margin $\mathcal{M}(x,t)/C_{\text{self}}$, which is precisely the Bekenstein–Nyquist Saturation Functional $\Sigma(x,t) = 1 - \frac{\|F_{tx}(x,t)\|_{\text{HS}}}{\Lambda_{\text{Bek}}\nu_{\text{ML}}}$ (Phase 2). When $\Sigma > 0$ ($R_{\text{self}} < C_{\text{self}}$), updates execute in $\mathcal{O}(1)$ parallel depth (elliptic regime); when $\Sigma < 0$ ($R_{\text{self}} > C_{\text{self}}$), Axiom A4 forces overflowing modes into sequential phase winding in $\mathcal{H}_{\text{clock}}$ via the 90° cotangent rotation $\xi \mapsto i\xi$ (hyperbolic regime).

4. **Deduction 4 — Impedance-Matched Givens Rotor and Landau–Ginzburg–Higgs Potential (From A3 + A4):** On the 2×2 subspace spanning the overflowing Nyquist bond mode $|u_{b,\text{UV}}\rangle \in V_\Lambda$ and the active clock wire $|c_{j,k}\rangle \in V_\Lambda^\perp$, reflectionless unitarity at $|k| = \Lambda_{\text{Bek}}$ (Axiom A4) uniquely fixes the SU(2) Givens rotation angle $\tan\vartheta = \sqrt{\Delta k/\Lambda_{\text{Bek}}}$. Because transferring norm $|c|^2$ into $\mathcal{H}_{\text{clock}}$ reduces the spatial commutator overload linearly ($F(|c|) = F_0 - \kappa|c|^2$), the saturation functional relaxes as $\Sigma(|c|) = \Sigma_0 + 2\lambda|c|^2$, whose integral with respect to $|c|^2$ yields the Landau–Ginzburg–Higgs potential $V_{\text{eff}}(c) = \Sigma_0|c|^2 + \lambda|c|^4$ (Phase 4).

1.3 Axiomatic Equivalence: Emergent Dilation vs. Primitive Bipartite Bundle

Just as quantum mechanics admits mathematically equivalent formulations (the Schrödinger, Heisenberg, and path-integral pictures), the Bekenstein–Nyquist Conjecture can be axiomatized either by deriving $\mathcal{H}_{\text{clock}}$ via Stinespring dilation (Axioms A1–A4) or by taking the bipartite fiber bundle $\mathcal{H}_{\text{tot}} = \mathcal{H}_\Lambda \oplus \mathcal{H}_{\text{clock}}$ as a primitive geometric structure equipped with minimal horizon gating:

- **Axiom B1 (Global Bundle Unitarity):** Composite evolution $U_{\text{tot}}(\Delta t) \in \text{U}(\mathcal{H}_\Lambda \oplus \mathcal{H}_{\text{clock}})$ conserves total norm $\|u\|_2^2 + \sum_{j,k} |c_{j,k}|^2 = \text{const}$.

- **Axiom B2 (Sub-Nyquist Dormancy / Horizon Gating):** The inter-bundle coupling $(I - P_\Lambda) U_{\text{tot}} P_\Lambda$ vanishes wherever $\|F_{tx}(x,t)\|_{\text{HS}} \leq \Lambda_{\text{Bek}}\nu_{\text{ML}}$ ($\Sigma(x,t) \geq 0$).

- **Axiom B3 (Impedance-Matched Overflow Serialization):** Wherever $\Sigma(x,t) < 0$, overflowing ultraviolet modes in $\mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$ rotate into $\mathcal{H}_{\text{clock}}$ via sequential rank-1 SU(2) Givens gates at $\tan\vartheta = \sqrt{\Delta k/\Lambda_{\text{Bek}}}$, costing one Margolus–Levitin tick $\Delta\tau_{\text{ML}} = \pi/\nu_{\text{ML}}$ per gate.

Proposition 1.1 (Theorem of Axiomatic Equivalence). *The Emergent Stinespring Dilation Axioms (A1–A4) and the Primitive Bipartite Bundle Axioms (B1–B3) are operationally equivalent on all spatial observables in V_Λ and induce identical sequential depth lower bounds $\mathcal{D}(\gamma)$, horizon metrics $\Sigma(x,t)$, and effective potentials $V_{eff}(c)$. Moreover, taking $\mathcal{H}_{clock}$ as a persistent co-moving bundle (B1–B3) naturally accommodates multi-step phase memory ($K = 64$ worldline clock slots in Milestone 4) and homogeneous vacuum expectation values ($\langle c \rangle_{vac} \neq 0$).*

Table 2: Three Complementary Perspectives on the Axiomatic Architecture of the Conjecture.

Axiomatic Lens	**Core Mathematical Statement**	**Ontological Meaning of a Shock Wave**	**Ontological Meaning of Mass / Clock Depth**
1. Operator-Algebraic / Homological Lens (Mathematicians & Lean 4)	Non-commutativity of nonlinear evolution and bandlimitation: $[P_\Lambda, U_{\mathrm{nl}}] \neq 0$	Localized spike in the commutator obstruction $\mathcal{A}_\Lambda[u] = \|[P_\Lambda, [M, L]]\|_{\mathrm{HS}}$	Minimal Stinespring dilation rank $D \geq \mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$ in $V_\Lambda^\perp$
2. Information-Theoretic / Queueing Lens (Computer Scientists)	Zero-drop channel buffering with capacity $C_{\mathrm{self}} = \Lambda_{\mathrm{Bek}} \nu_{\mathrm{ML}}$ and $U^\dagger U = I$	Local channel overload ($\mathcal{M} < 0 \iff \Sigma < 0$) where flux $R_{\mathrm{self}} > C_{\mathrm{self}}$	Sequential queueing lag $m_j \Delta\tau_{\mathrm{ML}}$ in local worldline clock memory $\mathcal{C}_j$
3. Geometric / Equivalence Principle Lens (Relativistic & Particle Physicists)	Covariant bandlimit invariance on complexified cotangent bundle $T^*\mathcal{M}_\mathbb{C}$	Sonic/Nyquist horizon ($\Sigma(x,t) = 0$) where effective metric $g_{00} \leftrightarrow \Sigma$ flips sign	Higgs/Kaluza–Klein condensate $\|c_{\mathrm{eq}}\|^2 = -\Sigma_0/(2\lambda)$ coiled in internal fiber

1.4 Analytical Proof Architecture for the Nyquist–Lax Trichotomy

To prove that spatial parallelism breaks down at a shock front under Axioms A1, A3, and A4, notice that "local unitarity" purely within the spatial Hilbert space $\mathcal{H}_\Lambda$ combines two distinct operational requirements: norm conservation ($\|\Phi_{\Delta t}(u)\|_2 = \|u\|_2$) and bounded-fan-in spatial locality ($\mathcal{O}(1)$ parallel circuit depth within stencil radius r_0). Theorem 1.2 proves that at a super-critical shock front, these two halves of spatial local unitarity cannot coexist with the Bekenstein bandlimit (Axiom A1), forcing the relaxation of $\mathcal{O}(1)$ parallel depth into sequential time-stepping governed by the Margolus–Levitin rate limit (Axiom A2).

Theorem 1.2 (Nyquist–Lax Trichotomy / Aliasing–Unitarity No-Go Lemma). *Any single-step spatial update operator $\Phi_{\Delta t} : \mathcal{H}_\Lambda \to \mathcal{H}_\Lambda$ acting across a shock locus where $\mathcal{A}_\Lambda[u] = \|[P_\Lambda, [M(u), L(u)]]\|_{HS} > \Theta_{DSW}$ must violate at least one of the following three operational postulates:*

1. ***Postulate 1 — Strict Spatial Bandwidth (Bekenstein Bound, Axiom A1):*** $\mathrm{supp}(\hat{u}) \subseteq [-\Lambda_{Bek}, \Lambda_{Bek}]$ *without dynamically allocating sub-Planckian spatial lattice nodes.*
2. ***Postulate 2 — Norm Unitarity / Information Conservation (First Half of Axiom A4):*** $\|\Phi_{\Delta t}(u)\|_2 = \|u\|_2$*, forbidding artificial viscosity, WENO truncation, or Landauer erasure of ultraviolet modes.*
3. ***Postulate 3 — Constant Parallel Depth / Bounded-Fan-In Spatial Locality (Second Half of Local Unitarity):*** $\Phi_{\Delta t}$ *is computable in circuit depth $\mathcal{O}(1)$ purely within $\mathcal{H}_\Lambda$ with a spatial stencil fan-in bounded by a fixed sub-Nyquist radius r_0.*

Proof Architecture. Assume Postulates 1, 2, and 3 hold simultaneously on $\mathcal{H}_\Lambda$.

1. **Step 1 (Balian–Low and Donoho–Stark Uncertainty Barrier):** By the quantitative Balian–Low theorem and the Donoho–Stark uncertainty inequality ($|T| \cdot |\Omega| \geq 1 - (\epsilon_T + \epsilon_\Omega)^2$), a shock discontinuity compressed to spatial width $\delta x \sim \pi/\Lambda_{\text{Bek}}$ concentrates $\mathcal{O}(1)$ spectral weight at the Brillouin boundary $|k| \in [\Lambda_{\text{Bek}}/2, \Lambda_{\text{Bek}}]$, generating a non-zero inter-band leakage operator $A = (I - P_\Lambda)U_{\text{nl}}P_\Lambda : \mathcal{H}_\Lambda \to \mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$.

2. **Step 2 (Unitary Defect Identity):** Decomposing the unconstrained unitary propagator $U_{\text{nl}} = \exp(\Delta t M(u))$ on $\mathcal{H}_\Lambda \oplus \mathcal{H}_\Lambda^\perp$ yields $(P_\Lambda U_{\text{nl}} P_\Lambda)^\dagger (P_\Lambda U_{\text{nl}} P_\Lambda) = P_\Lambda - A^\dagger A$. Thus, any bandlimited update $\Phi_{\Delta t} = P_\Lambda U_{\text{nl}} P_\Lambda$ dissipates norm $\|\Phi_{\Delta t} u\|_2^2 = \|u\|_2^2 - \|Au\|_2^2 < \|u\|_2^2$ (violating Postulate 2) whenever $Au \neq 0$.

3. **Step 3 (GNVW Chiral Index Obstruction and Group Velocity Reversal):** By the Gross–Nesme–Vogts–Werner (GNVW) index theorem for 1D Quantum Cellular Automata, any strictly local constant-depth unitary circuit restricted purely to $\mathcal{H}_\Lambda$ (Postulate 3) has a trivial inter-subspace flow index between $\mathcal{H}_\Lambda$ and $\mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$. Folding the chiral ultraviolet flux back into $[-\Lambda_{\text{Bek}}, \Lambda_{\text{Bek}}]$ via Umklapp aliasing ($k \mapsto k - 2\Lambda_{\text{Bek}}$) reverses the group velocity $v_g = \partial\omega/\partial k < 0$ at $|k| = \Lambda_{\text{Bek}}$ and produces non-physical backward dispersive ringing. Preserving both Postulates 1 and 2 therefore requires Stinespring dilation into an orthogonal register $\mathcal{H}_{\text{clock}}$ across sequential depth $D \geq \text{rank}(A)$ (violating Postulate 3) ticked at the Margolus–Levitin rate limit ν_{ML} (Axiom A2).

□

2 Microlocal Analysis of the 90° Causal Rotation (Phase 2)

2.1 The Bekenstein–Nyquist Saturation Functional and Mixed-Type PDEs

Once Phase 1 establishes that no constant-depth spatial operator on $\mathcal{H}_\Lambda$ can simultaneously preserve unitarity and bandwidth across a Lax curvature spike, Phase 2 constructs the unique dynamical escape mechanism forced by Axioms A1–A4: a microlocal signature transition that rotates overflowing spatial modes by 90° into sequential temporal phase degrees of freedom governed by ν_{ML}.

Definition 2.1 (Bekenstein–Nyquist Saturation Functional)**.** The local normalized feasibility margin $\mathcal{M}(x,t)/C_{\text{self}}$ of a bandlimited field $u(x,t) \in \mathcal{H}_\Lambda$ relative to the spacetime information-processing ceiling $\theta_{\text{Nyq}} = \Lambda_{\text{Bek}} \nu_{\text{ML}}$ is defined by

$$\Sigma(x,t) \equiv \frac{C_{\text{self}} - R_{\text{self}}(x,t)}{C_{\text{self}}} = 1 - \frac{\|F_{tx}(x,t)\|_{\text{HS}}}{\Lambda_{\text{Bek}}\, \nu_{\text{ML}}}. \tag{6}$$

The effective bandlimited propagator is modeled by a second-order mixed-type operator P_{eff} on the cotangent bundle $T^*\mathcal{M}$ whose principal symbol takes the generalized Tricomi–Keldysh form:

$$\sigma_{\text{eff}}(x,t;\xi,\omega) = \omega^2 + \Sigma(x,t)\, c_0^2 \xi^2, \tag{7}$$

where (ξ, ω) are the cotangent fiber coordinates dual to (x,t) and c_0 is the characteristic bulk wave speed:

- **Sub-Nyquist Bulk ($\Sigma(x,t) > 0$):** Where $\|F_{tx}(x,t)\|_{\text{HS}} < \Lambda_{\text{Bek}}\nu_{\text{ML}}$, $\sigma_{\text{eff}} = \omega^2 + |\Sigma| c_0^2 \xi^2$ has positive-definite **elliptic** signature $(+,+)$, admitting parallel equilibration in $\mathcal{O}(1)$ circuit depth ($m_j = 0$).

- **Sonic or Nyquist Horizon ($\Sigma(x,t) = 0$):** Along the codimension-1 caustic locus where $\|F_{tx}(x,t)\|_{\text{HS}} = \Lambda_{\text{Bek}}\nu_{\text{ML}}$, spatial stiffness vanishes ($\sigma_{\text{eff}} = \omega^2$), forming a parabolic degeneracy surface analogous to a transonic sonic horizon or mixed-type gravitational collapse horizon.

- **Super-Nyquist Shock Worldline** ($\Sigma(x,t) < 0$)**:** Inside the shock core where $\|F_{tx}(x,t)\|_{\mathrm{HS}} > \Lambda_{\mathrm{Bek}}\nu_{\mathrm{ML}}$, the ξ^2 coefficient flips sign to yield $\sigma_{\mathrm{eff}} = \omega^2 - |\Sigma|c_0^2\xi^2$. In the complexified cotangent fiber $T^*_{(x,t)}\mathcal{M}_{\mathbb{C}}$, this sign reversal is equivalent to a 90° Wick-type rotation $\xi \mapsto i\xi$, rendering the symbol strictly **hyperbolic** with Lorentzian signature $(-,+)$ and real characteristic lightcones $\frac{dx}{dt} = \pm\sqrt{|\Sigma(x,t)|}\, c_0$.

Table 3: Mixed-Type PDE Universality Classes at the Nyquist Horizon $\Sigma(x,t) = 0$.

Mixed-Type Class	**Canonical Normal Form near** $x_\perp = 0$	**Principal Symbol** $\sigma(x_\perp; \xi_\perp, \omega)$	**Characteristic Geometry at Horizon** $x_\perp = 0$	**Physical Role in Bekenstein–Nyquist Shocks**
Tricomi Type	$\partial_t^2 u + x_\perp c_0^2 \partial_{x_\perp}^2 u = 0$	$-\omega^2 - x_\perp c_0^2 \xi_\perp^2$	Characteristics $\frac{dx_\perp}{dt} = \pm c_0\sqrt{-x_\perp}$ meet the horizon **orthogonally** in a semi-cubic cusp and reflect	Governs transverse **Rankine–Hugoniot phase reflections** across the shock zone boundaries
Keldysh Type	$x_\perp \partial_t^2 u + c_0^2 \partial_{x_\perp}^2 u = 0$	$-x_\perp \omega^2 - c_0^2 \xi_\perp^2$	Characteristics approach the horizon **tangentially** (asymptotically) without escaping into the bulk	Governs **worldline trapping** of overflowing UV modes along the shock trajectory

2.2 Complex Coordinate Continuation and Internal Clock Winding

Across the turning point $\Sigma(x,t) = 0$, applying complex coordinate continuation $x \mapsto x + ic_{\mathrm{eff}}\tau$ paired with the 90° cotangent rotation $\Delta k \mapsto -i\Delta k$ maps an overflowing spatial mode $k = \Lambda_{\mathrm{Bek}} + \Delta k$ ($\Delta k > 0$) into sequential temporal phase winding along the internal worldline clock fiber $\mathcal{C}_j$:

$$\exp(ikx) = \exp(i\Lambda_{\mathrm{Bek}}x) \cdot \exp(i\Delta k\, x) \xrightarrow{x\mapsto x+ic_{\mathrm{eff}}\tau,\ \Delta k\mapsto -i\Delta k} \exp(i\Lambda_{\mathrm{Bek}}x) \cdot \exp(-i\omega_{\mathrm{sub}}\tau), \quad (8)$$

where the internal sub-stepping frequency is $\omega_{\mathrm{sub}} = c_{\mathrm{eff}}\Delta k \leq \nu_{\mathrm{ML}}$.

3 Complexity Theory and Sequential Depth Lower Bounds (Phase 3)

3.1 The Sub-Stepping Depth Theorem and Holographic Shock Complexity

Phase 3 translates the hyperbolic characteristic cones generated in the $\Sigma(x,t) < 0$ shock core into rigorous lower bounds on Boolean and unitary circuit depth, linking the PDE signature transition directly to the separation between parallel complexity (NC) and sequential polynomial time (P).

Theorem 3.1 (Sub-Stepping Depth Theorem)**.** *For any bounded-precision unitary simulator evolving a field configuration along a shock worldline γ, the minimum sequential circuit depth $\mathcal{D}(\gamma)$ is bounded below by the line integral of the projection-curvature commutator anomaly:*

$$\mathcal{D}(\gamma) \geq \frac{1}{\pi\nu_{ML}} \int_\gamma \|[P_\Lambda, [M(u), L(u)]]\|_{HS}\, ds. \quad (9)$$

This theorem exhibits a direct structural isomorphism with the **Complexity = Action (CA)** and **Shock Wave Geometry** framework in holographic quantum gravity (Stanford,

Susskind, Shenker, Maldacena, Brown, Roberts, Swingle, and Zhao). In holographic shock wave geometries, injecting shock waves into a bulk spacetime stretches the Wheeler–DeWitt (WDW) causal patch at a late-time rate governed by Lloyd's bound, $\frac{d\mathcal{C}}{dt} \leq \frac{2M}{\pi\hbar}$, while saturating the Maldacena–Shenker–Stanford (MSS) chaos bound $\lambda_L \leq \frac{2\pi k_B T}{\hbar}$. Here, each unit of integrated Hilbert–Schmidt commutator norm $\pi\nu_{\mathrm{ML}}$ requires at least one orthogonal Margolus–Levitin clock cycle to resolve unitarily. The proof strategy proceeds via **two-party communication complexity** across the moving shock cut $x_s(t)$ partitioning $\mathcal{H}_\Lambda$ into left ($\mathcal{H}_L$) and right ($\mathcal{H}_R$) Paley–Wiener subspaces: computing K alternating non-commuting Rankine–Hugoniot unitary phase reflections $R_K R_{K-1} \cdots R_1$ across the cut requires $\Omega(K)$ sequential message-passing rounds, forbidding compression into polylogarithmic NC depth $\mathcal{O}(\log^k K)$.

3.2 P-Completeness of Shock Worldline Networks and the Monotone Planar Circuit Trap

Table 4: Circuit Value Problem (CVP) Variants and Implications for Shock Worldline Networks.

CVP Variant	Topology	Gate Set	Complexity Class	Shock Network Implication
General CVP	Arbitrary DAG (crossings allowed)	$\{\wedge, \vee, \neg\}$	P-complete	Requires crossover gadgets in $1+1$D multi-species or $2+1$D collisions
Monotone CVP (MCVP)	Arbitrary DAG (crossings allowed)	$\{\wedge, \vee\}$ only	P-complete (Goldschlager 1977)	Cannot be embedded in strictly planar $1+1$D single-species flow
Planar CVP (PCVP)	Planar DAG (genus $g = 0$)	$\{\wedge, \vee, \neg\}$	P-complete (Goldschlager 1977)	**Target for Phase 3.2:** Planar shock intersections + unitary π-rotor negation ($\neg$)
Monotone Planar CVP (MPCVP)	Planar DAG (genus $g = 0$)	$\{\wedge, \vee\}$ only	In NC^3 ($\mathrm{LogDCFL} \subseteq \mathrm{NC}^2$ stratified)	**The MPCVP Trap:** Dissipative monotone shock merging is parallelizable in $\mathcal{O}(\log^3 N)$ depth

4 Computational Verification on Bandlimited Hydrodynamic Lattices (Phase 4)

4.1 The $1+1$D Unitary Nyquist Rotor Lattice (Experiment H.3 / Milestone 2)

To validate Phases 1 and 2 on a nonlinear hydrodynamic PDE without artificial viscosity or spectral aliasing, Experiment H.3 (`toy_model_2_nyquist_rotor_pde.py`) constructs the **Unitary Nyquist Rotor Lattice** in IEEE 754 `float64`/`complex128` precision on a periodic $1+1$D ring $x \in [-\pi, \pi)$ of $N_x = 2\Lambda_{\mathrm{Bek}}$ spatial nodes with fundamental Nyquist spacing $\Delta x = \pi/\Lambda_{\mathrm{Bek}}$. Each spatial site $j \in \{0, \ldots, N_x - 1\}$ is augmented with an M-level local worldline clock register $\mathcal{C}_j = \mathrm{span}\{|c_{j,1}\rangle, \ldots, |c_{j,M}\rangle\}$, yielding the composite Hilbert space $\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_\Lambda \oplus \bigoplus_{j=0}^{N_x-1} \mathcal{C}_j$. The solver evolves the inviscid Burgers conservation law $\partial_t u + \partial_x(\frac{1}{2}u^2) = 0$ via four strictly unitary, state-driven components:

1. **Translation-Invariant $\mathcal{H}_{2\Lambda}$ Analytic-Signal Spillover Sensor:** At each step, $u \in \mathcal{H}_\Lambda$ is zero-padded onto a $2\times$-oversampled Paley–Wiener grid $N_{\mathrm{fine}} = 4\Lambda_{\mathrm{Bek}}$ to evaluate $w_{\mathrm{fine}} =$

$u_{\text{fine}}\partial_x u_{\text{fine}}$. Retaining the positive super-Nyquist band $\Lambda_{\text{Bek}} \le k_{\text{fine}} < 2\Lambda_{\text{Bek}}$ defines the complex analytic signal:

$$w_{\text{analytic}}(x_{\text{fine}}) = 2\,\mathcal{F}^{-1}\big(\mathbf{1}_{\{\Lambda_{\text{Bek}} \le k_{\text{fine}} < 2\Lambda_{\text{Bek}}\}}\,\mathcal{F}(u_{\text{fine}}\partial_x u_{\text{fine}})\big)\,. \tag{10}$$

For a bandlimited step of jump Δu, $|w_{\text{analytic}}|$ scales organically as $c_\infty \Lambda_{\text{Bek}}(\Delta u)^2$ with $c_\infty \approx 0.264$ ($17.630, 34.580, 68.369$ at $\Lambda_{\text{Bek}} = 64, 128, 256$), determining $m_j = \lceil F_j/\theta_{\text{Nyq}} \rceil$.

2. **Skew-Symmetric Cayley Transport with Galilean-Invariant Delay:** Decomposing $u = \bar{u} + \delta u$ applies the queueing delay $\tau_j = m_j \Delta\tau_{\text{ML}}/\Delta t_{\text{ref}}$ strictly to the wave-frame fluctuation δu_j:

$$w_{\text{eff},j} = \frac{2}{3}\left(\bar{u}_j + \frac{u_j - \bar{u}_j}{1 + \gamma_{\text{eff}}\,\tilde{\tau}_j}\right), \qquad U_{\text{spatial}}(\Delta t) = \left(I + \frac{\Delta t}{2}A(w_{\text{eff}})\right)^{-1}\left(I - \frac{\Delta t}{2}A(w_{\text{eff}})\right), \tag{11}$$

where $A(w)^T = -A(w)$, guaranteeing $U_{\text{spatial}}^T U_{\text{spatial}} = I$ to machine precision ($\sim 10^{-16}$).

3. **5-Bond Unitary Haar–Givens Rotor Sweep:** Each active shock node executes m_j sequential 2×2 unitary Givens rotations $G_{j,k}(\vartheta_{j,k}, \phi_j) \in \mathrm{SU}(2)$ coupling the 2-site Haar Nyquist difference mode $u_{b,\text{UV}} = (u_{b_0} - u_{b_1})/\sqrt{2}$ to a single 1D clock wire $c_{j,\text{slot}} \in \mathcal{C}_j$ at the impedance-matched angle $\tan\vartheta = \sqrt{1/\Lambda_{\text{Bek}}}$.

4.2 Comparative Shock-Capturing Benchmarks and Poisson–Euler RG Duality

Table 5: Milestone 2 Comparative Benchmark Results ($\Lambda_{\text{Bek}} = 64, N_x = 128$).

Scenario	Solver Architecture	Max Norm Drift	Spatial $\|u\|_2^2$	Clock $\sum\|c\|^2$	Final $TV(u)$	Overshoot	Speed s (s_{RH})
Riemann Step	**Unitary Nyquist Rotor**	**1.245e-15**	32.638543	1.611907	1.6616	2.612e-02	0.5727 (0.6000)
Riemann Step	Dissipative Godunov	5.536e-02	32.354491	0.000000	1.5942	0.000e+00	0.6106 (0.6000)
Riemann Step	Naive Galerkin Spectral	3.615e-07	34.250438	0.000000	10.1806	1.767e-01	0.6080 (0.6000)
Sine Wave $-\sin(x)$	**Unitary Nyquist Rotor**	**6.661e-16**	58.267303	5.732697	4.1198	2.638e-02	0.0004 (0.0000)
Sine Wave $-\sin(x)$	Dissipative Godunov	1.120e-01	56.831485	0.000000	3.8780	0.000e+00	0.0000 (0.0000)
Sine Wave $-\sin(x)$	Naive Galerkin Spectral	4.590e-08	63.999997	0.000000	33.9979	3.446e-01	0.0000 (0.0000)

Because the impedance-matched Givens coupling scales as $\sin^2\vartheta = \frac{1}{\Lambda_{\text{Bek}}+1} \sim \Lambda_{\text{Bek}}^{-1}$ while the spectral spillover sub-step count at a shock scales as $m_j = c(\Delta u)\Lambda_{\text{Bek}}$, the per-macro-step ultraviolet energy absorption converges to a cutoff-independent Poisson–Euler renormalization-group fixed point:

$$\mathcal{T}_{\text{UV}}(\Lambda_{\text{Bek}}) = 1 - (\cos^2\vartheta)^{m_j} = 1 - \left(1 - \frac{1}{\Lambda_{\text{Bek}}+1}\right)^{c(\Delta u)\,\Lambda_{\text{Bek}}} \xrightarrow{\Lambda_{\text{Bek}}\to\infty} 1 - e^{-c(\Delta u)}. \tag{12}$$

4.3 Spacetime Causal DAG Critical-Path Scaling and Asymptotic Tightness (Experiment H.4 / Milestone 3)

Experiment H.4 (`toy_model_3_dag_complexity.py`) dynamically constructs the spacetime causal dependency DAG $G = (V, E)$ across bandwidth cutoffs $\Lambda_{\text{Bek}} \in \{64, 128, 256\}$ ($T_{\text{final}} = 1.60, \Delta t = 0.01$) for (a) a single-shock sine wave $u_0(x) = -\sin(x)$ ($t^* = 1.00$) and (b) a 4-mode random multi-shock gas ($t^* = 0.80$). At each macro-step Δt, m_j is partitioned across n_{sub} Cayley sub-steps via $m_{\text{sub},s} = \lfloor (m_j + s)/n_{\text{sub}} \rfloor$ so that the cumulative physical Givens rotations match the DAG rotor vertices 1-to-1:

$$\text{total_physical_givens_rotations} = \sum_{n=1}^{N_{\text{steps}}} \sum_{j=0}^{N_x - 1} m_j(t_n) = |V(T)| - (N_{\text{steps}} + 1)N_x. \tag{13}$$

Table 6: Milestone 3 Spacetime Causal DAG Critical-Path Scaling and Theorem Tightness.

Initial Condition	Λ_{Bek}	Pre-Shock	Final depth(G)	Excess D_{excess}	Bound $\mathcal{B}_{\text{thm}}$	Tightness	Work $\lvert V\rvert$	Rotations $\sum m_j$	Hyp. Slope	Max Drift
Single-Shock $-\sin(x)$	64	50	1014	854	816.19	**1.0463**	27086	6478	1637.04	6.661e-16
Single-Shock $-\sin(x)$	128	50	2244	2084	2052.35	**1.0154**	59121	17905	3903.70	1.332e-15
Single-Shock $-\sin(x)$	256	50	4429	4269	4240.79	**1.0067**	138963	56531	7933.33	1.554e-15
Multi-Shock Gas	64	40	541	381	329.83	**1.1551**	25545	4937	609.46	4.152e-16
Multi-Shock Gas	128	40	1014	854	813.46	**1.0498**	55041	13825	1243.24	1.107e-15
Multi-Shock Gas	256	40	1735	1575	1536.71	**1.0249**	120925	38493	2210.81	8.304e-16

4.4 The Bandwidth–Saturation Equivalence Principle and Holographic Bulk Extrusion

The analytical and numerical structure of Phases 2–5 establishes an operational **Bandwidth–Saturation Equivalence Principle**:

1. **Informational Horizon Geometry and Proper-Time Dilation ($\Sigma(x,t)$ as an Effective Metric Component g_{00}):** $\Sigma(x,t) = 1 - \frac{\|F_{tx}(x,t)\|_{\text{HS}}}{\Lambda_{\text{Bek}}\nu_{\text{ML}}}$ enters $\sigma_{\text{eff}}(x,t;\xi,\omega) = \omega^2 + \Sigma(x,t)c_0^2\xi^2$ in exact structural analogy to the Schwarzschild temporal metric component $g_{00}(r) = 1 - \frac{2GM}{c^2 r}$, inducing a proper-time queueing lag $\Delta\tau_{\text{lag}}(\gamma) = \mathcal{D}(\gamma)\Delta\tau_{\text{ML}}$ across $\Sigma < 0$.

2. **Inertial Mass as Sequential Queueing Drag:** In the Galilean-invariant effective transport velocity $w_{\text{eff},j} = \frac{2}{3}\big(\bar{u}_j + \frac{\delta u_j}{1+\gamma_{\text{eff}}\tilde{\tau}_j}\big)$ with $\tau_j = m_j\Delta\tau_{\text{ML}}/\Delta t_{\text{ref}}$, the factor $M_{\text{inert},j} \equiv 1+\gamma_{\text{eff}}\tilde{\tau}_j$ acts as a state-dependent inertial mass multiplier decelerating the over-steepened wave crest to the Rankine–Hugoniot shock speed $s = [f(u)]/[u]$.

3. **Rest Mass as Coiled Ultraviolet Momentum in the Clock Fiber:** In the stationary sine-wave shock benchmark ($u_0(x) = -\sin(x)$ at $T = 1.35 > t^* = 1.00$), the shock worldline sits macroscopically at rest at $x = 0$ ($s_{\text{RH}} = 0.0000$, measured speed 0.0004) while storing $\mathcal{M}_{\text{clock}} = \sum_{j,k}|c_{j,k}|^2 = 5.732697$ of ultraviolet L^2 norm in the stationary clock register $\mathcal{C}_{j_{\text{shock}}}$ ticking at $\omega_{\text{sub}} = c_{\text{eff}}\Delta k \le \nu_{\text{ML}}$.

4.4.1 Holographic Bulk Extrusion, Ryu–Takayanagi Min-Cut Wire Counting, and Effective g_{00} Horizon Geometry

Combining the Bekenstein–Nyquist Saturation Functional $\Sigma(x,t)$ of Phase 2 with the machine-checked Stinespring dilation bound of Phase 5 (`UnitaryBandwidthDepth.lean`) reveals a dynamical mechanism in which both an emergent orthogonal bulk/fiber dimension ($\mathcal{H}_{\text{clock}}$) and an effective horizon metric ($g_{00} \leftrightarrow \Sigma(x,t)$) are extruded by ultraviolet spectral overflow on the spatial boundary $\mathcal{H}_\Lambda$.

Corollary 4.1 (Boundary Overflow Drives Bulk Depth and Horizon Geometry)**.** *Let $u(x,t) \in \mathcal{H}_\Lambda = PW_{\Lambda_{Bek}}(\mathbb{R})$ evolve on a $1+1D$ bandlimited spatial lattice with Nyquist cutoff Λ_{Bek}, Margolus–Levitin transition ceiling ν_{ML}, and composite Hilbert space $\mathcal{H}_{tot} = \mathcal{H}_\Lambda \oplus \bigoplus_{j=0}^{N_x-1}\mathcal{C}_j$. Then:*

1. ***Dynamical Extrusion of the Orthogonal Fiber:*** *In the sub-Nyquist elliptic bulk ($\Sigma(x,t) > 0$, $F_j \le \theta_{Nyq} = \Lambda_{Bek}\nu_{ML}$), the spectral spillover into $\mathcal{H}_{2\Lambda} \ominus \mathcal{H}_\Lambda$ vanishes exponentially, leaving the internal clock fiber unexcited ($m_j = 0$) and the circuit depth flat ($\mathcal{O}(1)$). When nonlinear convective steepening breaches the Nyquist horizon ($\Sigma(x,t) < 0$, $F_j > \theta_{Nyq}$), the complex coordinate continuation $x \mapsto x + ic_{eff}\tau$ paired with the 90° cotangent rotation $\Delta k \mapsto -i\Delta k$ rotates overflowing boundary modes $\Delta k = k - \Lambda_{Bek} > 0$ into sequential temporal phase winding $\exp(-i\omega_{sub}\tau)$ along $\mathcal{C}_j$, dynamically extruding an internal fiber of sequential depth $m_j = \lceil F_j/\theta_{Nyq}\rceil \propto \Lambda_{Bek}(\Delta u)^2$ localized along the $1D$ shock worldline γ.*

2. ***Operator-Algebraic Ryu–Takayanagi / HRT Min-Cut Bound:*** *Each local Haar–Givens update decomposes a lattice bond (u_j, u_{j+1}) into its coarse-grained mean $\bar{u}_j = (u_j + u_{j+1})/\sqrt{2}$ and Nyquist difference mode $u_{j,UV} = (u_j - u_{j+1})/\sqrt{2}$, acting as a local multiscale wavelet disentangler (analogous to a MERA tensor network) rotated into sequential Margolus–Levitin proper time. By Theorem* `unitary_bandwidth_depth_tradeoff`*, any unitary factorization $U_D = W_D \cdots W_1$ whose elementary gates W_k couple $\mathcal{H}_\Lambda$ to at most a single 1D clock wire $C_k \subset \mathcal{H}_\Lambda^\perp$ ($\dim C_k = 1$) must sever a minimum number of clock wires along the worldline cut bounded by half the commutator rank:*

$$\begin{aligned} \operatorname{rank}\big([P_\Lambda, U_{nl}]\big) = 2r \implies D \geq r = \operatorname{rank}\big((I - P_\Lambda)U_{nl}P_\Lambda\big), \\ \mathcal{D}(\gamma) \geq \frac{1}{\pi \nu_{ML}} \int_\gamma \|[P_\Lambda, [M(u), L(u)]]\|_{HS}\, ds. \end{aligned} \tag{14}$$

3. ***Apparent Shock Entropy and Effective g_{00} Metric Warping:*** *Because total evolution on $\mathcal{H}_{tot}$ is strictly unitary ($\|u\|_2^2 + \sum_{j,k} |c_{j,k}|^2 = \text{const}$), macroscopic hydrodynamic entropy production is the entanglement/norm leakage $\mathcal{M}_{clock} = \sum_{j,k} |c_{j,k}|^2$ obtained by tracing out the worldline clock fiber $\mathcal{H}_{clock}$ across γ. Under impedance matching $\sin^2\vartheta = (\Lambda_{Bek}+1)^{-1}$, the per-step leakage along γ converges to the scale-invariant fixed point $1 - (\cos^2\vartheta)^{m_j} \to 1 - e^{-c(\Delta u)}$, while the sequential queueing delay $\Delta\tau_j = m_j \Delta\tau_{ML}$ warps the effective principal symbol $\sigma_{eff}(x,t;\xi,\omega) = \omega^2 + \Sigma(x,t)c_0^2\xi^2$ in exact correspondence with the temporal metric component $g_{00} \leftrightarrow \Sigma(x,t) = 1 - \frac{\|F_{tx}\|_{HS}}{\Lambda_{Bek}\nu_{ML}}$.*

4.5 Two-Shock Collision Interferometry, Coherent Clock-Mass Annihilation, and the Ginzburg–Landau–Higgs Duality (Milestone 4 & Phase 3.2)

4.5.1 Evasion of the Monotone Planar Circuit Trap via Orthogonal Two-Mode Haar Clock Readback

To demonstrate that intersecting 1 + 1D shock worldlines physically realize the non-monotone logical negation ($\neg$) gate required to evade the NC^3 collapse of MPCVP, Milestone 4 (`toy_model_4_shock_collision_gate.py`) replaces the write-only pointer advance with a **co-moving K-slot resonant worldline clock register** ($K = 64$, indexed directly by slot = (clock_ptr + k_{sub}) mod 64). On each active bond (b_0, b_1), `execute_bidirectional_givens_collision` transforms (u_{b_0}, u_{b_1}) into the orthonormal 2-site Haar wavelet basis:

$$\bar{u}_{\text{in}} = \frac{u_{b_0} + u_{b_1}}{\sqrt{2}} \quad \text{(smooth local mean)}, \qquad u_{\text{UV, in}} = \frac{u_{b_0} - u_{b_1}}{\sqrt{2}} \quad \text{(Nyquist difference)}, \tag{15}$$

and executes a three-stage orthogonal update on the 3-mode subspace $(\bar{u}_b, u_{b,\text{UV}}, c_{\text{slot}})$:

1. **Stage 1 — Unitary UV Stabilization ($u_{b,\mathbf{UV}} \leftrightarrow c_{\mathbf{slot}}$):** Rotates the overflowing Nyquist difference mode $u_{\text{UV, in}} > 0$ into the worldline clock slot c_{slot} by $\vartheta_1 = \min\big(\vartheta,\, 0.85 \arctan(u_{\text{UV, in}} / \max(0, c_{\text{in}}))\big)$:

$$\begin{pmatrix} u_{\text{UV, out}} \\ c_{\text{mid}} \end{pmatrix} = \begin{pmatrix} \cos\vartheta_1 & -\sin\vartheta_1 \\ \sin\vartheta_1 & \cos\vartheta_1 \end{pmatrix} \begin{pmatrix} u_{\text{UV, in}} \\ c_{\text{in}} \end{pmatrix}. \tag{16}$$

2. **Stage 2 — Phase-Coherent Collision Readback into the Smooth Mean Mode ($\bar{u}_b \leftrightarrow c_{\mathbf{slot}}$):** When the state-driven shock separation crosses $\Delta x_{\text{sep}} = x_2 - x_1 \leq 2.4\Delta x$, the relative rotor phase $\Delta\phi = \phi_2 - \phi_1$ governs $\gamma_{\text{phase}}(\Delta\phi) = \operatorname{sgn}(\sin(\Delta\phi/2)) \sin^2(\Delta\phi/2)$ and $\theta_{\text{rec}} = \min\big(\theta_{\text{cap}},\, 0.40\gamma_{\text{phase}} \arctan(c_{\text{mid}}/\bar{u}_{\text{in}})\big)$, rotating stored clock condensate c_{mid} back into the smooth local mean mode $\bar{u}_{\text{out}}$ without exciting $u_{\text{UV, out}}$:

$$\begin{pmatrix} \bar{u}_{\text{out}} \\ c_{\text{out}} \end{pmatrix} = \begin{pmatrix} \cos\theta_{\text{rec}} & \sin\theta_{\text{rec}} \\ -\sin\theta_{\text{rec}} & \cos\theta_{\text{rec}} \end{pmatrix} \begin{pmatrix} \bar{u}_{\text{in}} \\ c_{\text{mid}} \end{pmatrix}. \tag{17}$$

3. **Stage 3 — Inverse Orthogonal Haar Transformation:** Maps $(\bar{u}_{\text{out}}, u_{\text{UV, out}})$ back to (u_{b_0}, u_{b_1}):

$$\begin{aligned} u_{b_0}^{\text{out}} &= u_{b_0}^{\text{in}} + \frac{\bar{u}_{\text{out}} - \bar{u}_{\text{in}}}{\sqrt{2}} + \frac{u_{\text{UV, out}} - u_{\text{UV, in}}}{\sqrt{2}}, \\ u_{b_1}^{\text{out}} &= u_{b_1}^{\text{in}} + \frac{\bar{u}_{\text{out}} - \bar{u}_{\text{in}}}{\sqrt{2}} - \frac{u_{\text{UV, out}} - u_{\text{UV, in}}}{\sqrt{2}}. \end{aligned} \tag{18}$$

4.5.2 Experimental Verification: Gate Truth Table, Coalescence Kinematics, and Ramsey Interference Fringes

Table 7: Milestone 4 Suite B Two-Shock Collision & Coalescence Benchmark ($T_{\text{final}} = 1.40, \Lambda_{\text{Bek}} = 64$).

Collision Solver / Regime	$\Delta\phi$	Clock Mass $\mathcal{M}_{\text{clock}}(T)$	Spatial $\|u(T)\|_2^2$	Max $\|u\|$	Final $TV(u)$	Max Drift	Speed s_{merged}	Rotations $\sum m_j$
Unitary Rotor — In-Phase	0	3.017201	49.695025	1.2433	2.8108	**8.089e-16**	0.6769 (-3.30%)	1449 == 1449
Unitary Rotor — Anti-Phase	π	**0.396683**	**52.315543**	**1.3157**	**2.8143**	**8.089e-16**	0.7573 (+8.19%)	1858 == 1858
Dissipative Godunov / WENO	—	0.000000	49.155968	1.2000	2.0000	6.746e-02	0.7000	0

Table 8: Milestone 4 Suite C Shock Mach–Zehnder / Ramsey Interference Fringe ($\Delta\phi \in [0, 2\pi]$).

$\Delta\phi$	$\Delta\phi$ (deg)	$\mathcal{M}_{\text{clock}}(T, \Delta\phi)$	$\|u(T, \Delta\phi)\|_2^2$	$\|u\|_2^2 + \mathcal{M}_{\text{clock}}$	Max $\|u(x,T)\|_\infty$	Final $TV(u)$	$\sum m_j$	Max Drift
0	0°	3.017201	49.695025	52.712226	1.2433	2.8108	1449	8.089e-16
$\pi/4$	45°	2.167711	50.544516	52.712226	1.3082	3.8474	1382	1.483e-15
$\pi/2$	90°	1.120733	51.591494	52.712226	1.3046	2.9885	1499	8.089e-16
$3\pi/4$	135°	0.687753	52.024473	52.712226	1.3337	3.7520	1443	8.089e-16
π	180°	**0.396683**	**52.315543**	52.712226	**1.3157**	**2.8143**	**1858**	8.089e-16
$5\pi/4$	225°	0.687753	52.024473	52.712226	1.3337	3.7520	1443	8.089e-16
$3\pi/2$	270°	1.120733	51.591494	52.712226	1.3046	2.9885	1499	8.089e-16
$7\pi/4$	315°	2.167711	50.544516	52.712226	1.3082	3.8474	1382	1.483e-15
2π	360°	3.017201	49.695025	52.712226	1.2433	2.8108	1449	8.089e-16

4.5.3 Duality Between the Bekenstein–Nyquist Clock Fiber and the Landau–Ginzburg–Higgs Mechanism

Because Stage 1 transfers ultraviolet energy $|c|^2$ from $u_{b,\text{UV}}$ into the clock fiber, the active commutator defect decreases linearly as $F(|c|) = F_0 - \kappa|c|^2$, shifting the instantaneous saturation functional to

$$\Sigma(|c|) = 1 - \frac{F_0 - \kappa|c|^2}{\Lambda_{\text{Bek}}\,\nu_{\text{ML}}} = \Sigma_0 + 2\lambda|c|^2, \qquad \lambda \equiv \frac{\kappa}{2\Lambda_{\text{Bek}}\nu_{\text{ML}}} > 0. \tag{19}$$

Integrating $\frac{\partial V_{\text{eff}}}{\partial(|c|^2)} = \Sigma(|c|)$ yields the effective Landau–Ginzburg–Higgs Mexican-hat potential governing the clock condensate:

$$V_{\text{eff}}(c) = \Sigma_0|c|^2 + \lambda|c|^4 = \underbrace{\left(1 - \frac{F_0(x,t)}{\Lambda_{\text{Bek}}\,\nu_{\text{ML}}}\right)}_{\mu^2(x,t) \equiv \Sigma_0(x,t)} |c|^2 + \lambda|c|^4, \tag{20}$$

where the Higgs boson mass $m_H^2 c^4 \propto \frac{1}{2}\frac{\partial^2 V_{\text{eff}}}{\partial|c|^2}\big|_{|c|=|c_{\text{eq}}|} = -2\Sigma_0(\hbar\nu_{\text{ML}})^2$ represents the elastic restoring stiffness of the Sonic/Nyquist Horizon $\Sigma = 0$.

Table 9: Four-Pillar Structural Duality Between the Landau–Ginzburg–Higgs Mechanism and the Bekenstein–Nyquist Clock Fiber.

Structural Pillar	**Landau–Ginzburg–Higgs Formulation**	**Bekenstein–Nyquist Worldline Formulation**	**Mathematical Duality**
1. Origin of Auxiliary Fiber (Phase 5)	Perturbative unitarity restoration: scalar doublet Φ absorbs $(E/\Lambda)^2$ longitudinal growth at the Lee–Quigg–Thacker cutoff	Stinespring dilation (`UnitaryBandwidthDepth.lean`): $P_\Lambda U_{\mathrm{nl}} P_\Lambda$ requires $D \geq \mathrm{rank}([P_\Lambda, U_{\mathrm{nl}}])/2$ clock wires in $V_\Lambda^\perp$	**Sz.-Nagy / Stinespring Duality:** Higgs field Φ is the continuum effective representation of the UV clock complement $V_\Lambda^\perp = (I - P_\Lambda)H$
2. Condensation Trigger (Phase 2)	Spontaneous symmetry breaking: quadratic coefficient μ^2 in $V(\Phi) = \mu^2\lvert\Phi\rvert^2 + \lambda\lvert\Phi\rvert^4$ flips from $\mu^2 > 0$ to $\mu^2 < 0$	Bandwidth saturation: $\Sigma_0(x,t) = 1 - \frac{F_0}{\Lambda_{\mathrm{Bek}}\nu_{\mathrm{ML}}}$ flips from $\Sigma_0 > 0$ (elliptic) to $\Sigma_0 < 0$ (hyperbolic shock core)	**Horizon Criticality:** $\mu^2(x,t) \equiv \Sigma_0(x,t)$; condensing $\lvert c_{\mathrm{eq}}\rvert^2 = -\Sigma_0/(2\lambda)$ restores horizon equilibrium $\Sigma(\lvert c_{\mathrm{eq}}\rvert) = 0$
3. Momentum-Space Locus (Phases 2 & 4)	Infrared zero-mode condensate ($k \to 0$): rest mass $m_0c^2 = \hbar\omega_0$ is a homogeneous internal Compton oscillation $\exp(-i\omega_0\tau)$	Ultraviolet Brillouin overflow ($\lvert k\rvert > \Lambda_{\mathrm{Bek}}$): $x \mapsto x + ic_{\mathrm{eff}}\tau$ rotates $\Delta k = k - \Lambda_{\mathrm{Bek}} > 0$ into internal clock winding $\exp(-i\omega_{\mathrm{sub}}\tau)$	**UV/IR Fibre Duality:** Shifting momentum origin by Λ_{Bek} and rotating 90° maps boundary UV overflow into fiber IR oscillation
4. Worldline Dynamics (Phases 3 & 4)	Relativistic worldline action: massive particle accumulates proper-time phase $\frac{1}{\hbar}\int_\gamma mc^2\, d\tau$ via chiral Yukawa mixing $\psi_L \leftrightarrow \psi_R$	Sub-Stepping Depth Theorem: shock worldline γ executes $\mathcal{D}(\gamma) \geq \frac{1}{\pi\nu_{\mathrm{ML}}}\int_\gamma \lVert[P_\Lambda,[M,L]]\rVert_{\mathrm{HS}}\, ds$ sequential Givens rotations	**Action–Complexity Duality:** Relativistic mass action along γ is dual to sequential Margolus–Levitin gate depth along the causal DAG critical path

5 Machine-Checked Formal Verification in Lean 4 (Phase 5, Phase A & Phase B)

5.1 Core Bandwidth–Depth Tradeoff (`UnitaryBandwidthDepth.lean`)

In `UnitaryBandwidthDepth.lean` (verified against Lean 4 v4.33.1 and Mathlib with zero `sorry` axioms and zero compiler warnings), we formalize the finite-dimensional Hilbert space $H \cong \mathbb{C}^{d+m}$ indexed by the canonical block sum type `BlockIdx d m := Fin d ⊕ Fin m`. The orthogonal bandlimit projector onto the d-dimensional low-frequency Bekenstein subspace V_Λ and its m-dimensional ultraviolet complement $V_\Lambda^\perp$ ($m = N - d$) are defined via `Matrix.fromBlocks` as $P_\Lambda = \mathrm{diag}(I_d, 0_m)$ (`bandlimitProj d m`) and $I - P_\Lambda = \mathrm{diag}(0_d, I_m)$ (`uvProj d m`):

$$
\begin{aligned}
P_\Lambda &= \begin{pmatrix} I_d & 0 \\ 0 & 0 \end{pmatrix}, \qquad U_{\mathrm{nl}} = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 0 \\ U_{21} & 0 \end{pmatrix}, \\
B &= \begin{pmatrix} 0 & U_{12} \\ 0 & 0 \end{pmatrix}, \qquad [P_\Lambda, U_{\mathrm{nl}}] = B - A = \begin{pmatrix} 0 & U_{12} \\ -U_{21} & 0 \end{pmatrix}.
\end{aligned} \tag{21}
$$

The formalization establishes four machine-checked theorems:

1. **Single-Wire Clock Support Implies Rank-One Leakage (`rank_forwardLeakage_le_one_of_isElementaryClockGate`):** An elementary clock gate $W \in \mathrm{U}(d+m)$ satisfies `IsElementaryClockGate d m c W` if its forward leakage $(I - P_\Lambda)WP_\Lambda$ vanishes on all ultraviolet rows $\mathrm{inr}(i) \neq \mathrm{inr}(c)$ outside a single 1D clock wire $c \in \mathrm{Fin}(m)$. Factoring the non-zero row through `Fin 1` proves $\mathrm{rank}((I - P_\Lambda)WP_\Lambda) \leq 1$.

2. **One-Sided Leakage Subadditivity (`forwardLeakage_mul` & `rank_forwardLeakage_prod_le_length`):** For any idempotent projector $P^2 = P$ and any list of unitary gates $L : \mathrm{List}(\mathrm{Matrix}\ n\ n\ \mathbb{C})$, the algebraic identity

$$(I-P)(WU)P = \big((I-P)WP\big)(PUP) + \big((I-P)W(I-P)\big)\big((I-P)UP\big) \tag{22}$$

combined with rank subadditivity (`Matrix.rank_add_le`) and induction over L proves that if every $W \in L$ has $\mathrm{rank}((I-P)WP) \le 1$, then $\mathrm{rank}((I-P)(\prod L)P) \le \mathrm{length}(L) = D$.

3. **Commutator Rank Duality (`forwardLeakage_rank_eq_reverseLeakage_rank` & `commutator_rank_eq_two_mul_forwardLeakage_rank`):** Using the unitary block Gram identities $U_{11}^\dagger U_{11} + U_{21}^\dagger U_{21} = I_d$ and $U_{11}U_{11}^\dagger + U_{12}U_{12}^\dagger = I_d$ together with `Matrix.rank_conjTranspose_mul_self` ($\mathrm{rank}(M^\dagger M) = \mathrm{rank}(M)$), the formalization proves that forward and reverse Nyquist leakage have equal rank ($\mathrm{rank}(A) = \mathrm{rank}(B) = r$) and reside in orthogonal subspaces, yielding $\mathrm{rank}([P_\Lambda, U]) = 2r$.

4. **Isometric and Block-Compression Bandwidth–Depth Tradeoffs (`unitary_bandwidth_depth_tradeoff_compression` & `unitary_bandwidth_depth_tradeoff_clock_wires`):** For any target nonlinear unitary $U_{\mathrm{nl}} \in \mathrm{U}(d+m)$ simulated by a sequential circuit $U_D = \prod_{k=1}^{D} W_k$ of elementary 1D clock gates, both exact isometric action agreement ($U_D P_\Lambda = U_{\mathrm{nl}} P_\Lambda$) and the weaker block-compression dilation condition ($P_\Lambda U_D P_\Lambda = P_\Lambda U_{\mathrm{nl}} P_\Lambda$) force $\mathrm{rank}((I-P_\Lambda)U_D P_\Lambda) = \mathrm{rank}((I-P_\Lambda)U_{\mathrm{nl}} P_\Lambda) = r$, establishing the unconditional lower bound:

$$D = \mathrm{length}(L) \ge r = \frac{1}{2}\mathrm{rank}\big([P_\Lambda, U_{\mathrm{nl}}]\big). \tag{23}$$

5.2 Extended Physics Formalizations and Upstream `Physlib` / `QuantumInfo` Bridges

To connect the abstract operator-algebraic core of `UnitaryBandwidthDepth.lean` to the physical mechanisms of Phases 2–4 and to the community Lean 4 physics ecosystem (`leanprover-community/physlib`, which bundles `Physlib` and `QuantumInfo`), five additional machine-checked Lean 4 modules have been constructed and verified with zero `sorry` axioms and zero compiler warnings:

1. **Concrete SU(2) Givens Rotors, Pauli Blockade, and Tripartite Collision Router (`NyquistRotorGates.lean`):** Instantiates `IsElementaryClockGate` with explicit 2×2 SU(2) Givens rotors $G_\phi(\vartheta)$ (`givensRotor_mem_unitaryGroup`, `givensRotor_isElementaryClockGate`). Machine-checks constructive in-phase clock-mass creation $G_0(\vartheta_2)G_0(\vartheta_1) = G_0(\vartheta_1 + \vartheta_2)$ (`givensRotor_in_phase_mul`), anti-phase coherent clock-mass annihilation $G_\pi(\vartheta)G_0(\vartheta) = I_2$ realizing the non-monotone logical negation ($\neg$) gate required to evade the Monotone Planar CVP NC^3 trap (`givensRotor_anti_phase_mul_eq_one`), the general Ramsey collision fringe entry $(1+\zeta)\sin\vartheta\cos\vartheta$ (`givensRotor_ramsey_fringe_entry`), Pauli saturation reflection at $\vartheta = \pi/2$ (`pauli_saturation_reflection`), the rank-nullity Pauli exclusion bound $\dim\ker((I-P_\Lambda)WP_\Lambda) \ge (d+m) - 1$ (`pauli_exclusion_blockade_ker_finrank`), strict unitarity and zero UV-spike excitation of the Milestone 4 2-stage tripartite Haar router on $(\bar{u}_b, u_{b,\mathrm{UV}}, c_{\mathrm{slot}})$ (`stage2MeanReadback_preserves_uv_mode`, `milestone4TripartiteRouter_mem_unitaryGroup`), and the spatial–clock Gram partition of unity $(PUP)^\dagger(PUP) + A^\dagger A = P$ (`spatial_compression_add_clock_leakage_eq_proj`).

2. **Landau–Ginzburg–Higgs Horizon Criticality Duality (`NyquistHiggsDuality.lean`):** Formalizes the Bekenstein–Nyquist saturation functional

$\Sigma_0 = 1 - F_0/\theta_{\mathrm{Nyq}}$ (`saturationFunctional`), linear commutator defect relaxation $\Sigma(|c|^2) = \Sigma_0 + 2\lambda|c|^2$ (`saturation_linear_relaxation`), and the effective Mexican-hat potential $V_{\mathrm{eff}}(|c|^2) = \Sigma_0|c|^2 + \lambda|c|^4$ (`nyquistEffectivePotential`). Proves completion of the square (`HiggsPotential.complete_square`), global lower boundedness (`HiggsPotential.pos_l_lower_bound`), unique vacuum minimization $|c_*|^2 = 0$ in the sub-Nyquist bulk $\Sigma_0 > 0$ (`subNyquist_unique_minimum_zero`), spontaneous clock-condensate formation $|c_{\mathrm{eq}}|^2 = -\Sigma_0/(2\lambda) > 0$ in the super-Nyquist shock core $\Sigma_0 < 0$ (`superNyquist_condensate_minimum`), exact restoration of the sonic horizon $\Sigma(|c_{\mathrm{eq}}|^2) = 0$ (`superNyquist_horizon_restoration`), and cutoff-scale invariance of λ (`quarticCoupling_cutoff_independent`).

3. **Cayley Integrator Unitarity and Gauge-Covariant Clock Parallel Transport (`NyquistGaugeTransport.lean`):** Proves that for any skew-Hermitian operator $D^\dagger = -D$ and left resolvent $R(I + D) = I$, the Crank–Nicolson / Cayley transform $U = R(I - D)$ is unconditionally unitary (`cayleyTransform_mem_unitaryGroup`). Establishes skew-Hermiticity of both the split-form Burgers advection matrix (`splitFormAdvection_skewHermitian`) and the gauge-covariant clock parallel-transport generator $D^{\mathrm{cov}}_{i,j} = w_{i,j}\sigma_{i,j}U_{i,j}$ (`covariantClockAdvection_skewHermitian`), unitary gauge equivariance $D^{\mathrm{cov}}(U') = G(\psi)D^{\mathrm{cov}}(U)G(\psi)^\dagger$ (`covariantClockAdvection_gauge_equivariant`), and exact local U(1) gauge invariance of two-shock Ramsey collision holonomy $U^{(1)}_{i,k}\overline{U^{(2)}_{i,k}}$ (`twoPathHolonomy_gauge_invariant`) and closed 3-site Wilson loops (`wilsonLoop3_gauge_invariant`).

4. **Direct Upstream `Physlib` Standard Model Higgs Bridge (`PhyslibHiggsBridge.lean`):** Imports `Physlib.Particles.StandardModel.HiggsBoson.Potential` directly from `leanprover-community/physlib` and bridges it with `NyquistHiggsDuality.lean` over smooth sections φ : `HiggsField` on 4D `SpaceTime`. Proves pointwise identity between upstream `StandardModel.HiggsField.Potential.toFun` and `nyquistEffectivePotential` under $\mu^2 = -\Sigma_0$ (`toPhyslibPotential_toFun_eq_nyquistEffectivePotential`) and directly invokes upstream `isMinOn_iff_field_of_$\mu$Sq_nonpos_l_pos` and `isMinOn_iff_field_of_$\mu$Sq_nonneg_l_pos` to certify the sub-Nyquist vacuum ($\varphi(x) = 0$), super-Nyquist clock condensate ($\|\varphi(x)\|_H^2 = -\Sigma_0/(2\lambda)$), and sonic-horizon restoration (`superNyquist_physlib_horizon_restoration`).

5. **Direct Upstream `QuantumInfo` CPTP Channel & Von Neumann Entropy Bridge (`QuantumInfoClockBridge.lean`):** Imports `QuantumInfo.Channels.CPTP` and `QuantumInfo.Entropy.VonNeumann` directly from `leanprover-community/physlib`. Certifies our SU(2) Givens rotor and Milestone 4 tripartite Haar router as upstream unitary `CPTPMap` channels (`givensRotorCPTP_isUnitary`, `milestone4TripartiteCPTP_isUnitary`), constructs the reduced spatial fluid channel $\Phi_\Lambda(\rho_\Lambda) = \mathrm{Tr}_C(U(\rho_\Lambda \otimes \rho_C)U^\dagger)$ via `CPTPMap.traceRight` (`reducedSpatialFluidChannel_tracePreserving`), proves Stinespring purification duality via `CPTPMap.purify_trace` (`spatialChannel_admits_unitary_purification`), and formalizes Section **??** by proving that for any pure joint fluid+clock state Ψ : `Ket`$(d_{\mathrm{spatial}} \times d_{\mathrm{clock}})$, global von Neumann entropy vanishes identically $S_{\mathrm{vN}}(|\Psi\rangle\langle\Psi|) = 0$ (`bipartite_pure_global_entropy_zero`), while the apparent macroscopic shock entropy of the reduced spatial fluid equals the entanglement entropy stored in the worldline clock fiber $S_{\mathrm{vN}}(\rho_\Lambda) = S_{\mathrm{vN}}(\rho_C) \geq 0$ (`spatial_shock_entropy_eq_clock_entropy`) with mutual information $I(\Lambda : C) = 2S_{\mathrm{vN}}(\rho_\Lambda)$ (`bipartite_pure_mutual_info_eq_two_mul_spatial_entropy`).

Table 10: Summary of the Six Machine-Checked Lean 4 Formalization Modules (Zero `sorry` Axioms).

Lean 4 Module	**Phase / Scope**	**Key Formalized Structures & Theorems**	**Monograph Section & Physical Role**
`UnitaryBandwidthDepth.lean`	Phase 5 (Core Mathlib)	`bandlimitProj`, `IsElementaryClockGate`, $\text{rank}([P_\Lambda, U_{\text{nl}}]) = 2r$, $D \geq r$	Sections 1, 3 & 5.1: Stinespring dilation & sequential clock-wire depth bound
`NyquistRotorGates.lean`	Phase A (Test A1)	`givensRotor`, $G_0(\vartheta_2)G_0(\vartheta_1) = G_0(\vartheta_1 + \vartheta_2)$, $G_\pi(\vartheta)G_0(\vartheta) = I_2$, `milestone4TripartiteRouter`	Sections ?? & 4.5: π-rotor $\neg$ gate, Pauli blockade, & UV-spike-free Haar readback
`NyquistHiggsDuality.lean`	Phase A (Test A2)	`saturationFunctional`, `nyquistEffectivePotential`, $\lvert c_{\text{eq}}\rvert^2 = -\Sigma_0/(2\lambda)$, $\Sigma(\lvert c_{\text{eq}}\rvert^2) = 0$	Sections ?? & 4.5: Spontaneous clock-condensate formation & sonic-horizon restoration
`NyquistGaugeTransport.lean`	Phase A (Test A3)	`cayleyTransform_mem_unitaryGroup`, `covariantClockAdvection`, `twoPathHolonomy_gauge_invariant`	Sections ?? & ??: Unconditional Cayley unitarity & U(1) Wilson holonomy invariance
`PhyslibHiggsBridge.lean`	Phase B (Upstream `Physlib`)	`toPhyslibPotential`, `subNyquist_physlib_isMinOn_iff_zero`, `superNyquist_physlib_isMinOn_iff_condensate`	Section 4.5: Direct bridge to `StandardModel.HiggsField.Potential` on 4D `SpaceTime`
`QuantumInfoClockBridge.lean`	Phase B (Upstream `QuantumInfo`)	`givensRotorCPTP`, `reducedSpatialFluidChannel`, `spatial_shock_entropy_eq_clock_entropy`	Sections ?? & ??: `CPTPMap` purification & $S_{\text{vN}}(\rho_\Lambda) = S_{\text{vN}}(\rho_C)$

6 Integrated Synthesis

Together, the Minimal Four-Axiom Formulation (Axioms A1–A4) and its equivalence with the Primitive Bipartite Bundle (Axioms B1–B3), the analytical proofs of Phases 1–3, the six machine-checked Lean 4 modules of Phase 5 and Phases A–B (`UnitaryBandwidthDepth.lean`, `NyquistRotorGates.lean`, `NyquistHiggsDuality.lean`, `NyquistGaugeTransport.lean`, `PhyslibHiggsBridge.lean`, and `QuantumInfoClockBridge.lean`), and the four progressive computational toy models of Phase 4 (`toy_model_1_unitary_leakage.py` through `toy_model_4_shock_collision_gate.py`, passing 45/45 automated unit, falsification, and anti-cheat assertions) establish a unified mathematical and computational foundation for the **Bekenstein–Nyquist Shock Worldline Conjecture**:

- **Unitary Dilation Lower Bound (Milestone 1 / Phase 5):** Any unitary simulation of a nonlinear operator across a bandlimit projector P_Λ using 1D clock wires requires an orthogonal clock fiber $V_\Lambda^\perp$ and sequential circuit depth $D \geq \text{rank}([P_\Lambda, U_{\text{nl}}])/2$.
- **Zero-Loss Unitary Shock Stabilization (Milestone 2 / Experiment H.3):** Rotating overflowing Nyquist bond modes into local worldline clock registers via impedance-matched U(2) Givens rotations ($\tan\vartheta = \sqrt{\Delta k/\Lambda_{\text{Bek}}}$) eliminates Gibbs ringing while conserving total Hilbert-space norm $\|u\|_2^2 + \sum |c|^2$ to $\sim 10^{-15}$ and propagating shocks at the Rankine–Hugoniot speed $s = [f(u)]/[u]$.
- **Emergent Complexity Phase Transition & Asymptotic Tightness (Milestone 3 / Experiment H.4):** Purely state-driven $\mathcal{H}_{2\Lambda}$ spectral spillover detection yields $\mathcal{O}(1)$ parallel circuit depth for $t < t^*$ and transitions at $t \approx t^*$ to linear sequential depth

$\Omega((T-t^*)\Lambda_{\text{Bek}})$ along 1D shock worldlines, converging to within 0.67% of the continuum Sub-Stepping Depth Theorem integral bound at $\Lambda_{\text{Bek}} = 256$.

- **Holographic Bulk Extrusion & Ryu–Takayanagi Min-Cut Correspondence (Subsection 4.4.1):** Ultraviolet Bekenstein–Nyquist overflow on the spatial boundary $\mathcal{H}_\Lambda$ dynamically extrudes sequential depth in the orthogonal clock fiber $\mathcal{H}_{\text{clock}}$ and warps the effective horizon metric $g_{00} \leftrightarrow \Sigma(x,t)$, providing an operator-algebraic realization of Ryu–Takayanagi min-cut wire counting along γ.
- **Coherent Clock-Mass Readback & Landau–Ginzburg–Higgs Duality (Milestone 4 / Phase 3.2):** Coupling a co-moving worldline clock register to the orthogonal 2-site Haar basis $(\bar{u}_b, u_{b,\text{UV}})$ realizes both the unitary π-rotor logical negation ($\neg$) gate required for Planar CVP P-completeness and an 86.85% coherent clock-mass annihilation fringe across $\Delta\phi \in [0, 2\pi]$.

References

[1] Primary Conjecture Blueprint, Bekenstein–Nyquist Research Program Archive.

[2] Physical Capacity Regions of Computation, Preprints.org (2025), 202509.0010.v1.

[3] Conditional Cosmological Recurrence in Finite Hilbert Spaces, *Universe* 12(1), 10.

[4] arXiv:2009.01340v2 [hep-th].

[5] Quantum gravity, information theory and the CMB, arXiv:1803.01483v1.

[6] Kempf, A., Information-theoretic natural ultraviolet cutoff for spacetime, arXiv:gr-qc/0310035v1.

[7] Locality and entanglement in bandlimited quantum field theory, *Phys. Rev. D* 92, 105022 (2015).

[8] Kempf, A., Covariant Information-Density Cutoff in Curved Space-Time, *Phys. Rev. Lett.* 92, 221301 (2004).

[9] Xuereb, J. (2026), On the inherent thermodynamics of quantum information processing, TU Wien.

[10] Detector Bounds or Spacetime Bounds? Operational Limits, *Int. J. Mod. Phys. D* (2026), 10.1142/S0218271826500574.

[11] From Seiberg–Witten Geometry to Categorical Integrability, ResearchGate 395704905.

[12] The Whitham Modulation Solution of the Complex Modified KdV, *Mathematics* 11(13), 2810.

[13] A systematic method for constructing time discretizations, arXiv:0906.3155.

[14] arXiv:1512.08668v2 [math.FA].

[15] A quantitative Balian–Low theorem, arXiv:1205.0163v1.

[16] Jansen, K., Lower Bound Frontiers in Arithmetical Circuit Complexity, University at Buffalo.

[17] Clifford Quantum Cellular Automata from Topological Phases, APS Journals.

[18] Walter, M., Approximate QCAs and a converse to the Lieb–Robinson bounds.

[19] Witteveen, F., Classifying unitary dynamics with approximate light cones in one dimension.

[20] Whitham modulation theory for generalized Whitham equations, UC Merced.

[21] Dispersive Hydrodynamics of Soliton Condensates for the Korteweg–de Vries Equation, PMC10509137.

[22] arXiv:2207.13074v2 [gr-qc].

[23] Gravitational collapse in Einstein–dilaton–Gauss–Bonnet gravity, ResearchGate 333332378.

[24] Brown, A. R., Roberts, D. A., Susskind, L., Swingle, B., & Zhao, Y., Complexity, action, and black holes, arXiv:1512.04993v2.

[25] Quantum Computational Complexity, arXiv:2110.14672v1.

[26] Holographic complexity and thermodynamics of AdS black holes, *Phys. Rev. D* 100, 026016.

[27] A Fully Quantum Algorithm for Hydrodynamic Lattice Gas Cellular Automata, arXiv:2310.07362v1.

[28] Evaluating Monotone Circuits on Cylinders, Planes and Tori, STACS 2006.

[29] Limaye, N., Mahajan, M., & Sarma, J., Upper Bounds for Monotone Planar Circuit Value.

[30] Evaluating Monotone Circuits on Surfaces, ECCC Report 2022/147.

[31] On Wave-Based Majority Gates with Cellular Automata, ResearchGate 355221563.

[32] Generalized Shannon Sampling Method reduces the Gibbs Overshoot in the Approximation of a Step Function, ResearchGate 228993443.

[33] `Mathlib.LinearAlgebra.Matrix.Rank`, Lean Prover Community Documentation.

[34] Quantum space-depth tradeoffs for coherent block encodings, arXiv:2607.01843v2.